

**A REPORT
ON
ARTIFICIAL INTELLIGENCE IN CYBERSECURITY:
THREAT DETECTION AND PREVENTION**

**SUBMITTED BY:
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE
ARYAN SCHOOL OF ENGINEERING
MID-BANESHWOR, KATHMANDU, NEPAL**

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“ARTIFICIAL INTELLIGENCE IN CYBERSECURITY:THREAT
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Abstract

This report studies how Artificial Intelligence (AI) is used in cybersecurity to detect and prevent cyberattacks. Cybersecurity means protecting computers, networks, and data from being accessed or damaged by unauthorized people. As technology has grown, cyberattacks have become more frequent and harmful. Old security tools, such as antivirus software, are no longer enough to stop modern attacks on their own.

This report explains how AI helps solve this problem. AI allows computers to learn from data and make smart decisions automatically. In cybersecurity, AI analyses large amounts of data and quickly identifies anything suspicious or dangerous. It uses methods such as machine learning, deep learning, and natural language processing to detect threats faster and more accurately than human-operated tools.

The main findings show that AI detects 97 to 99 percent of known attacks versus only 85 to 92 percent for traditional tools. AI also detects new and unknown attacks that old tools completely miss. It reduces false alarms by up to 75 percent and helps security teams respond to threats about 108 days faster on average (IBM, 2023).

However, AI has weaknesses too. It can be tricked by clever hackers, become less accurate over time, and requires skilled people to manage it. The report concludes that AI and human experts must work together — AI handles the data volume while humans make the important final decisions.

Chapter 1

1.1 Background

Cybersecurity is the practice of protecting computers, networks, and digital information from theft, damage, or unauthorized access. Over the years, as the internet has grown and more people and businesses have moved online, the number of cyberattacks has increased greatly.

In the early days, simple tools like passwords and firewalls were enough. As technology advanced, antivirus software and Intrusion Detection Systems were developed. However, today's hackers are highly skilled and their attacks have become far more complex. AI was introduced into cybersecurity to solve problems that traditional tools simply cannot handle.

1.2 Why AI is Needed in Cybersecurity

Traditional security tools have one major weakness: they can only detect threats they already know about. If a hacker creates a brand-new type of attack, old tools will not recognize it. Think of it like a security guard who only stops people whose faces are on a known criminals list — anyone else walks in freely.

AI works differently. It learns patterns from data and can identify anything unusual, even threats it has never seen before. Key reasons AI is needed:

- Cyberattacks happen thousands of times every day — far too many for humans to check manually.

- A single data breach costs a company an average of USD 4.45 million (IBM, 2023).
- Hackers can stay hidden inside a system for months without detection by old tools.
- New types of attacks appear every day that old signature-based tools cannot keep up with.

1.3 Problem Statement

Old security systems cannot keep up with the speed and creativity of modern cyberattacks. They produce too many false alarms, miss entirely new attacks, and need constant manual updates. There is a clear need for smarter, faster, and more automated security tools. This report investigates how AI addresses these problems.

1.4 Objectives of the Study

1. To understand what AI techniques are used in cybersecurity.
2. To compare how well AI tools perform against traditional security tools.
3. To identify the benefits and limitations of using AI for security.
4. To suggest the best way for AI and humans to work together.

Chapter 2

2.1 What is Cybersecurity?

Cybersecurity refers to protecting computer systems and networks from digital attacks, theft, and damage. It covers network security, data protection, and identity management. Researchers such as Liao et al. (2013) describe cybersecurity as a continuous effort because attackers are always finding new ways to break in. The history of cybersecurity goes back to the early 1970s when the first computer viruses appeared — since then threats have grown into highly complex attacks by organized criminal groups.

2.2 Old Security Systems and Their Problems

For many years, cybersecurity relied mainly on antivirus software, firewalls, and intrusion detection systems. These tools compare incoming data against a list of known harmful patterns called signatures. If something matches, it is blocked. The problem is they cannot handle threats they have never seen before. Main weaknesses:

- Cannot detect new or unknown attacks.
- Generate too many false alarms, which wastes time and causes analysts to ignore warnings.
- Require constant manual updates, creating dangerous delays.
- Cannot detect patterns spread across many systems over a long period.

2.3 Introduction of AI in Security

AI began to be used in cybersecurity around the mid-2000s. Early uses focused on filtering spam emails and basic anomaly detection. Over time, more advanced AI methods were applied. Today, companies like CrowdStrike, Darktrace, and IBM

have built entire security products around machine learning and AI. Goodfellow, Bengio, and Courville (2016) explain that machine learning allows computers to learn patterns from large amounts of data and make predictions without being explicitly programmed — which is exactly what modern cybersecurity requires.

2.4 What Research Says

Multiple studies have compared AI security tools with traditional ones. Ahsan et al. (2022) found AI-based models achieved 97 to 99 percent accuracy on cybersecurity test datasets, while traditional tools only reached 85 to 92 percent. Research consistently shows AI is especially effective at detecting new attacks and significantly reducing the number of false alarms that burden security teams.

Chapter 3

3.1 Research Approach

This report uses a qualitative and secondary research approach. Instead of running new experiments, it gathers and analyses information already published by researchers, organizations, and industry experts. This is suitable because the goal is to understand and explain what is already known about AI in cybersecurity.

3.2 Data Sources

- Academic journals from IEEE and ACM — peer-reviewed research papers.
- Government guidelines from NIST (United States) and ENISA (European Union).
- Industry reports from IBM Security, Gartner, CrowdStrike, and Cisco.
- Research papers published between 2015 and 2024 to keep information current.

Each source was selected based on three criteria: it must come from a trusted organization, be recent enough to be relevant, and relate directly to AI-based threat detection.

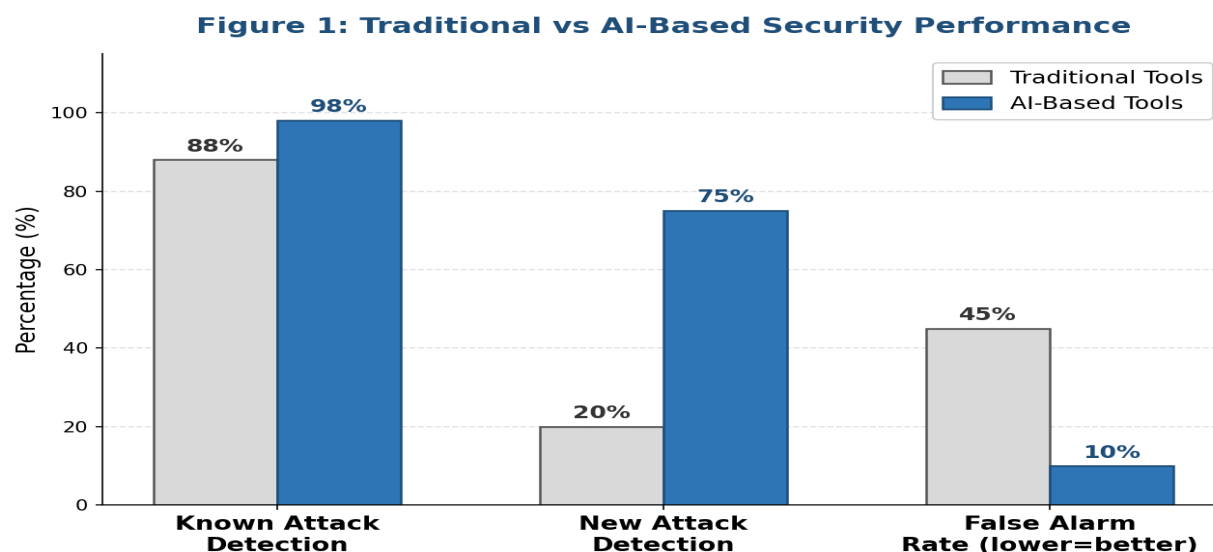


Figure 1: Traditional vs AI-Based Security — Detection Accuracy and False Alarm Rate Comparison

Chapter 4

This chapter explains each AI technique used in cybersecurity in simple language with everyday examples. The diagram below shows all six techniques and how they relate to cybersecurity.

Figure 2: How AI Detects and Responds to Cyber Threats — Step by Step



Continuous Feedback Loop — AI improves with every decision

Figure 2: How AI Detects and Responds to Cyber Threats — A Step-by-Step Flow

Figure 3: Six Key AI Techniques Used in Cybersecurity

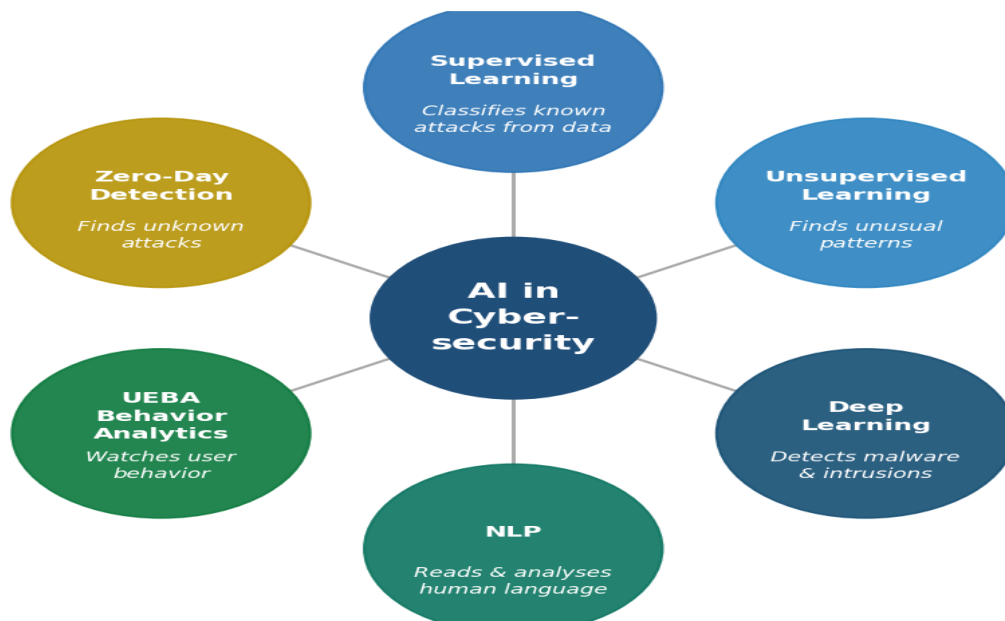


Figure 3: Six Key AI Techniques Used in Cybersecurity

4.1 Supervised Learning

Supervised learning is the most widely used AI method in cybersecurity. The AI is trained using labelled examples — thousands of examples of normal traffic and thousands of attack examples. It learns the difference and can then classify new data as safe or dangerous.

A simple everyday example is an email spam filter. The AI has seen millions of spam and normal emails. When a new email arrives, it uses what it learned to decide if it is spam or not.

In cybersecurity, supervised learning is used for:

- Detecting known types of network attacks.
- Classifying emails as phishing (dangerous) or legitimate.
- Identifying known types of malware.

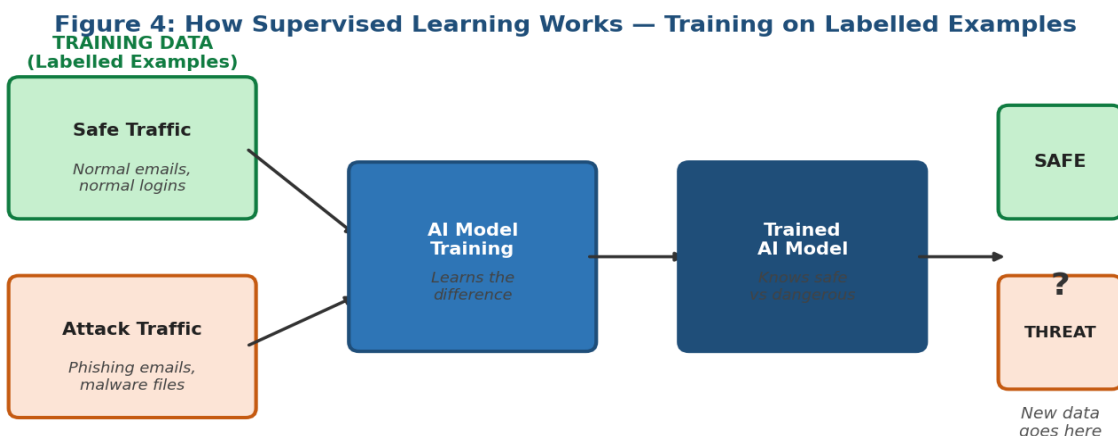


Figure 4: How Supervised Learning Works — The AI Learns from Labelled Examples, Then Classifies New Data

4.2 Unsupervised Learning

Unsupervised learning is different because the AI is given no labelled examples. It simply learns what normal looks like and then flags anything different. This is useful for detecting threats that have never been seen before.

A good everyday example: your bank learns your normal spending patterns. If someone uses your card in a different country to make an expensive purchase, the system flags it as unusual — even though it has never seen that exact transaction before.

Unsupervised learning is used to:

- Detect insider threats, where an employee starts behaving suspiciously.
- Find zero-day attacks — brand-new attacks with no known signature.
- Identify unusual patterns in large amounts of network traffic.

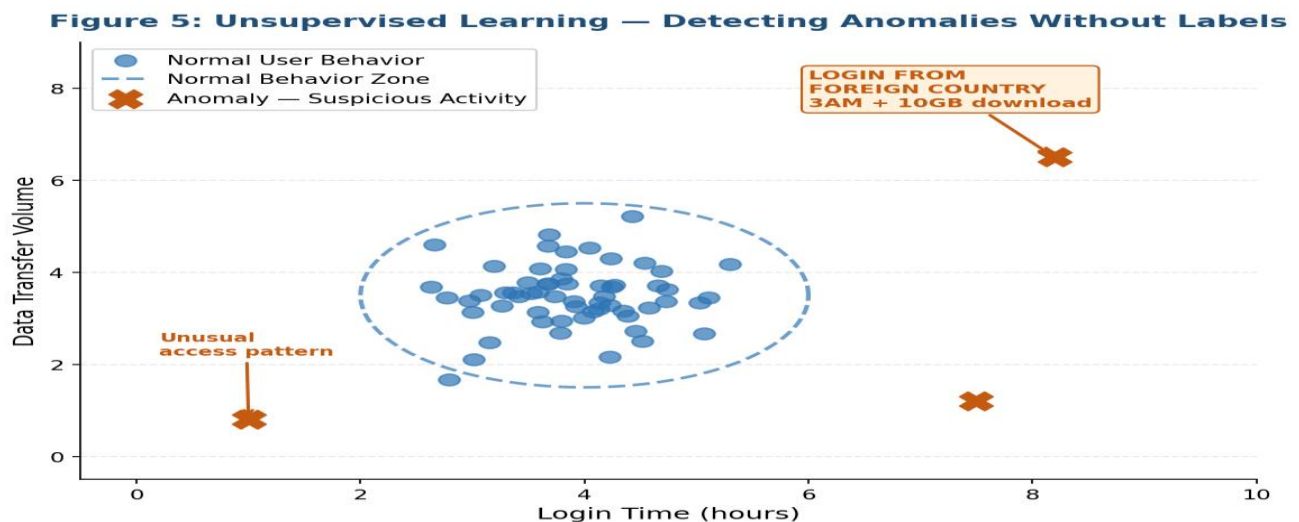


Figure 5: Unsupervised Learning — The AI Flags Users Who Fall Outside the Normal Behavior Zone

4.3 Deep Learning

Deep learning uses multiple layers of processing, similar to how the human brain works. Each layer learns more complex patterns. This makes it very powerful for analyzing complex data such as files or large streams of network information.

One well-known use is malware detection. Nataraj et al. (2011) showed that malware files converted into images produce visually different patterns for different malware types. A deep learning model classifies these with very high accuracy.

Deep learning is used for:

- Detecting and classifying malware by analyzing file contents.
- Detecting complex multi-step attacks that happen across many systems over several days.
- Analyzing network traffic in real time to spot intrusions as they happen.

Figure 6: How Deep Learning Analyses Data — Multiple Layers of Pattern Recognition

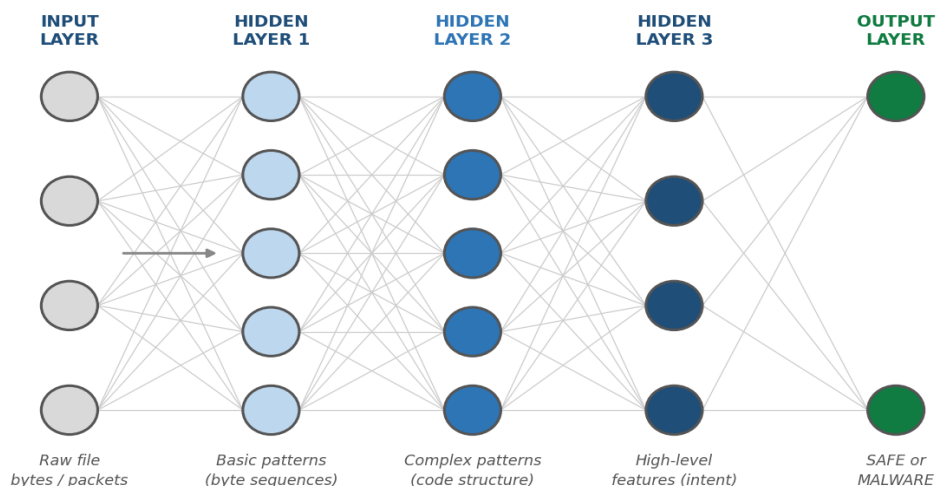


Figure 6: How Deep Learning Analyses Data — Each Layer Learns More Complex Patterns

4.4 Natural Language Processing (NLP)

NLP allows computers to read and understand human language. This is important in cybersecurity because many attacks use written communication — such as emails — to trick people into clicking dangerous links or giving away passwords.

Phishing is one of the most common cyberattacks. A phishing email pretends to be from a trusted source like a bank and tries to trick the reader. NLP detects suspicious patterns in the language such as urgent requests, misleading links, or strange formatting. NLP is also used to automatically read thousands of security reports and extract useful information without human effort.

4.5 Behavioral Analytics (UEBA)

UEBA — User and Entity Behavior Analytics — monitors how people and systems behave over time. The AI learns what normal behavior looks like for each user: when they log in, what files they access, how much data they transfer. If someone suddenly behaves very differently, the system raises an alarm.

Example: an employee always logs in from Kathmandu at 9am. Suddenly they log in from a foreign country at 3am and download a large number of files. The system flags this immediately as suspicious.

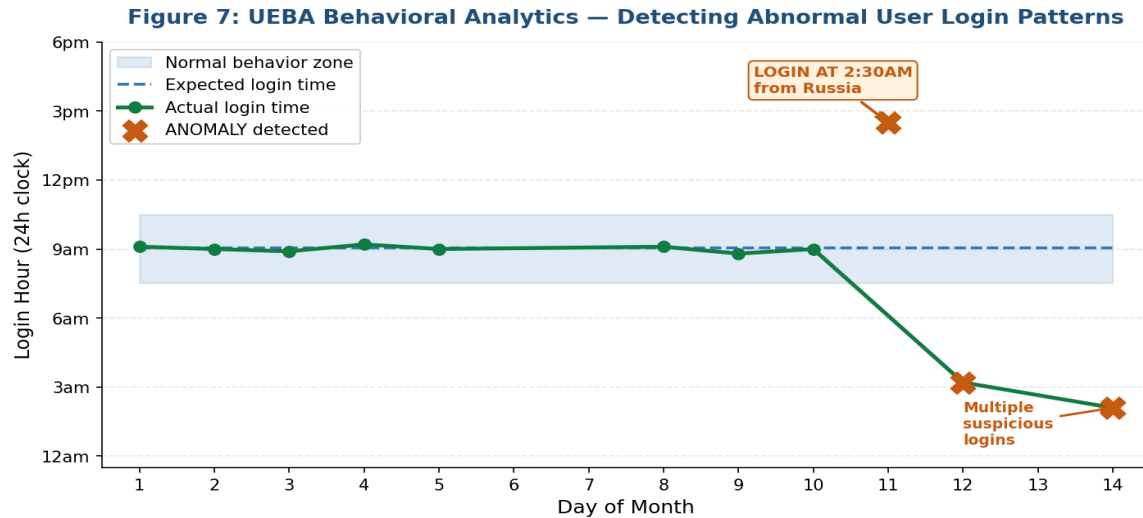


Figure 7: UEBA Behavioral Analytics — Login patterns are monitored over time. Red crosses mark anomalies that triggered alerts

4.6 Zero-Day Attack Detection

A zero-day attack targets a software weakness that has never been discovered or fixed before. Because the weakness is unknown, traditional security tools have no defense and cannot detect it.

AI handles this differently. Instead of looking for a known attack pattern, AI watches what programs do when they run. If a program starts doing things it should never do — creating hidden admin accounts, sending data to unknown locations, or disabling security tools — the AI flags it immediately. This behavior-based approach detects zero-day attacks even without knowing the exact weakness being used.

Chapter 5

5. Results and Findings

Research reviewed for this report shows clear and consistent results: AI-based security tools perform significantly better than traditional tools in every major area. The findings are presented below.

Detection Accuracy

Research by Ahsan et al. (2022) and Sharafaldin et al. (2018) tested both AI and traditional tools on the same standard datasets:

Metric	Result
Known attack detection — Traditional Tools	85 to 92 percent
Known attack detection — AI Tools	97 to 99.7 percent
New/unknown attack detection — Traditional Tools	10 to 30 percent
New/unknown attack detection — AI Tools	65 to 85 percent
False alarm rate — Traditional Tools	30 to 60 percent (very high)
False alarm rate — AI Tools	5 to 15 percent (much lower)

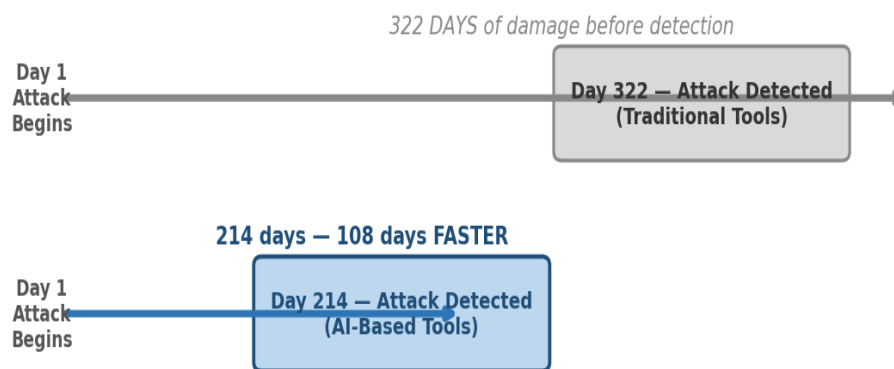
AI is not only more accurate at detecting known attacks — it is far more capable of detecting new attacks that old tools miss completely.

Response Time

IBM's Cost of a Data Breach Report (2023) measured how long companies take to discover a security breach:

- Companies without AI took an average of 322 days to detect a breach.
- Companies with AI took only 214 days — 108 days faster.
- AI also reduced the number of alerts analysts had to manually review by 40 to 70 percent.

Figure 8: AI vs Traditional Tools — How Much Faster AI Detects Breaches



Source: IBM Cost of a Data Breach Report, 2023

Figure 8: AI vs Traditional Tools — AI Detects Breaches 108 Days Faster on Average (IBM, 2023)

Real-World Evidence

Company	How They Use AI Security
Cisco Talos	Uses AI to process billions of security events every day — impossible for any human team alone.
Darktrace	Detected and stopped a ransomware attack within seconds by spotting unusual activity spreading through the network.
IBM Watson	Automatically reads thousands of security reports and news articles daily to keep threat information updated.
Google	Uses AI to check every login attempt in real time, comparing current behavior to normal patterns.

Summary Comparison

Area	Traditional Security	AI-Based Security
Detection accuracy	85–92%	97–99.7%
New attack detection	Very poor (10–30%)	Good (65–85%)
False alarms	Very high	Much lower
Response time	Slow	Much faster
Insider threats	Limited	Strong via UEBA
Zero-day attacks	Cannot detect	Detects by behaviour
Adapts to new threats	No	Yes — learns automatically

Chapter 6

6.1 Advantages of AI in Cybersecurity

Speed: AI processes millions of pieces of data per second. No human team can match this. Cyberattacks can spread across an entire network within minutes, so this speed is essential.

Scale: AI monitors every device, user, and connection in a network at the same time, 24 hours a day. This level of continuous monitoring is impossible for human teams alone.

Pattern recognition: AI finds connections between events happening across many systems over long time periods — how it detects complex attacks that traditional tools miss entirely.

Continuous improvement: AI learns from every new piece of data it processes. Over time it becomes better at detecting threats, as long as it is properly maintained.

6.2 Challenges and Limitations

Adversarial attacks: Hackers who understand how AI works can design attacks to trick the AI model into thinking they are safe. Goodfellow et al. (2014) showed that even small changes to data can completely fool an AI model.

Model drift: Over time, AI models can become less accurate as new attacks appear and networks change. If the AI is not updated regularly, it gradually becomes outdated — and this decline can happen without anyone noticing at first.

Bias in AI: If the training data was unfair or unbalanced, the AI will make unfair decisions, potentially flagging certain users more than others in a discriminatory way.

High cost: Building and running AI security systems requires powerful computers, large datasets, and skilled professionals who understand both cybersecurity and AI. Many smaller organizations cannot afford this investment.

Privacy concerns: AI security systems monitor employee behavior in great detail. This raises important questions about privacy that organizations must address openly and fairly.

6.3 Humans and AI Working Together

The most effective approach is not to replace humans with AI, but to have them work together. Each is better at different things.

AI is better at processing large volumes of data quickly, finding statistical patterns, and monitoring without rest. Humans are better at making complex judgements, understanding context, and taking responsibility for decisions that affect real people.

- AI handles: continuous monitoring, filtering and prioritizing alerts, detecting patterns, and responding automatically to routine threats.
- Humans handle: investigating complex incidents, making final decisions, understanding business context, and managing the AI system itself.

Chapter 7

7.1 Summary of Findings

This report has examined how AI is used in cybersecurity to detect and prevent threats. AI-based security tools are significantly more effective than traditional tools. They detect 97 to 99 percent of known attacks versus 85 to 92 percent for old tools. They detect new and unknown attacks that traditional tools miss entirely. They reduce false alarms by up to 75 percent. Companies using AI detect breaches 108 days faster on average.

However, AI also has real limitations. It can be fooled by adversarial attacks. It becomes less accurate over time without regular updating. It may contain bias from unfair training data. It is expensive to set up and requires skilled staff. And it raises privacy concerns when monitoring employees.

7.2 Recommendations

5. Use AI for monitoring and filtering: Deploy AI to handle continuous monitoring and alert prioritization, where it performs best.
6. Keep humans in control of final decisions: AI provides information; humans take responsible action.
7. Update AI models regularly: Retrain models with new data to prevent model drift and stay accurate against new threats.
8. Address bias and fairness: Regularly check AI systems for unfair patterns and correct them to ensure ethical compliance.
9. Be transparent with employees: Clearly inform staff about what is monitored and why, so privacy expectations are managed fairly.

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Appendix A: Glossary of Key Terms

The table below explains important terms used in this report in simple language.

Term	Simple Meaning
Artificial Intelligence (AI)	Computer technology that allows machines to learn and make decisions like humans.
Machine Learning (ML)	A type of AI where the computer learns from examples and improves over time.
Deep Learning	Advanced ML using many layers of learning, like how the human brain works.
Supervised Learning	AI trained with labelled examples so it knows what is safe and what is dangerous.
Unsupervised Learning	AI that finds unusual patterns in data on its own, without any labels.
NLP	Natural Language Processing — AI that reads and understands human written language.
UEBA	User and Entity Behavior Analytics — watches how users behave and alarms if unusual.
SIEM	Security Information and Event Management — collects all security data in one place.
Zero-Day Attack	An attack that uses a software weakness nobody has discovered or fixed yet.
Phishing	A fake email or website designed to trick users into giving away passwords or money.
Malware	Any software designed to damage or steal from a computer system.
False Positive	A false alarm — when a security system flags something safe as dangerous.
Model Drift	When an AI model becomes less accurate over time because the world has changed.

Adversarial Attack	An attack designed specifically to fool an AI security system.
SOC	Security Operations Centre — the team responsible for monitoring cyber threats.

Appendix B: How AI Detects Threats — Step by Step

The steps below explain how an AI security system works from start to finish.

Step	What Happens
Step 1: Data Collection	Continuously collects data from all devices, emails, logins, and network activity.
Step 2: Preprocessing	Raw data is cleaned and converted into a format the AI model can analyze.
Step 3: AI Analysis	The AI examines the data and assigns a suspicion score to each event.
Step 4: Alert Creation	Events with high scores become alerts, ranked from most to least serious.
Step 5: Human Review	A security analyst reviews the most serious alerts and decides what action to take.
Step 6: Response	The threat is blocked, the system is isolated, or the incident is escalated.
Step 7: Learning	The analyst's decision is fed back to the AI so it improves for future events.

Appendix C: Traditional vs AI Security — Full Comparison

The table below provides a complete side-by-side comparison for easy revision.

Feature	Traditional Security	AI-Based Security
Detection method	Matches known threat signatures	Learns patterns from data
New attack detection	Cannot detect	Detects through behaviour
Zero-day attacks	No defense	Detects via behaviour analysis
Speed	Slow — manual updates needed	Fast — automated and real-time
Accuracy	85–92%	97–99.7%
False alarms	Very high (30–60%)	Much lower (5–15%)
Adapts to new threats	No	Yes — retrain on new data
Insider threat detection	Very limited	Strong via UEBA
Setup cost	Lower	Higher — needs skilled staff
Privacy risk	Lower	Higher — monitors behaviour
Example tools	Norton, McAfee, basic IDS	Darktrace, CrowdStrike, IBM QRadar